



Nursing Practice Guideline

Chest Drain Management

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Nursing

Introduction

This guideline outlines the procedures and responsibilities of individuals and the health service organisation in relation to management of chest drain/s. It ensures the delivery of a uniform, integrated multidisciplinary approach with management that is consistent with best practice.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☒	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☐	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGOPHCG Mission & Values	☒	Staff Safety	☐
Other:	☐		

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Principles

To ensure optimal outcomes for patients with a chest drain, care needs to be provided by appropriately skilled staff who maintain close vigilance of the patient, the chest drain and drainage system; and are able to implement appropriate actions in the event that complications arise.¹⁻⁴

Chest drain insertion requires written consent as per North Metropolitan Health Service Consent to Treatment Policy, and in compliance with the WA Health Consent to Treatment Policy (MP 0175/22)

Observe the five moments for hand hygiene. Follow aseptic technique and use the risk assessment outlined in the SCGOPHCG Infection Prevention and Control Manual- Aseptic Technique Policy. This must include use of appropriate Personal Protective Equipment (PPE) to prevent contamination and mucosal or conjunctival splash injuries

Patient identification must be checked when matching patient's identity to care, therapy, and services.

1.0 General Information and Care

1.1 Medical Requirements

A Medical Officer should review the patient if there is:

- Bleeding from the insertion site¹
- A significant air leak from the insertion site
- Problems with recurrent kinking of the chest drain
- Signs of infection around the chest drain insertion site^{2, 4}
- Problems with fixation sutures²
- Subcutaneous emphysema^{1† ‡}

1.2 Safety Information

To ensure optimal outcomes for patients with a chest drain, care needs to be provided by appropriately skilled staff who maintain close vigilance of the patient, the chest drain and drainage system; and are able to implement appropriate actions in the event that complications arise.¹⁻⁴

Staff should be familiar with the basic principles and care information provided in this Section.

In general, chest drains for pneumothorax should not be clamped except briefly when replacing chest drainage bottle or chest drain tubing set.² Clamping may result in a tension pneumothorax or surgical emphysema.^{2,3,}

Prior to drain removal, some clinicians may request a 'Clamping Trial' to detect small air leaks not immediately obvious at the bedside. These patients must remain in the clinical area and be closely observed for signs of respiratory distress for the duration of the clamping trial.² Immediately unclamp if signs of clinical deterioration.^{2,3}

Patients with a chest drain to an underwater seal system must have a nurse escort familiar with chest drain management when leaving ward area.¹

A chest drain does **NOT** require clamping to transport patient.^{1,4}

1.3 Nursing Management

Principles of Chest Drain Management

- Chest drains are inserted through the intercostal space into the pleural space between the visceral and parietal pleura to enable evacuation of air and/or fluid from the pleural space to facilitate re-expansion of the lung.² (See Table 1 below for available types and sizes.)

Table 1. Chest Drain sizes and Indications

Chest Drain	Size	Insertion	Indications	Removal
Large Bore Intercostal Catheters (ICC)	18 – 36 French	Inserted intra-operatively or through an incision using blunt dissection. NB 18 French may also be inserted using a Seldinger technique.	Chest surgery or trauma for drainage of viscous pleural fluid and has - blood (haemothorax), - air and blood (haemopneumothorax), - pus (empyema) - lymphatic fluid (chylothorax).¹	Requires TWO nurses. (See Section 7)
Small Bore Pleural Drains	12 - 16 French	Inserted with guide wire and dilators using a Seldinger technique.	Drainage of air (pneumothorax) or drainage of less viscous pleural fluid. ¹	Requires ONE nurse. If closing suture present, TWO nurses are required. (See Section 7)
Indwelling Pleural Catheters (IPC):	16 French	Tunnelled chest drains incorporating a Dacron cuff; inserted using a combination of Seldinger and blunt dissection techniques guided by thoracic ultrasound.	Long-term management of malignant pleural effusions associated with mesothelioma or other lung malignancies. ⁵ IPC's can be capped and accessed intermittently to drain pleural fluid. ⁵	A surgical procedure performed by Respiratory Physicians only.

- A chest drain should be connected to a drainage system that contains a valve system to prevent fluid or air from entering the pleural space.^{1,3,6}
- Chest drainage systems consist of 3 components:
 - A drainage collection canister^{6,7}
 - A seal – most commonly an underwater seal (UWS), created by submerging the last 2cm of the drain tubing in water. This creates a low resistance one-way

valve that allows air and fluid to be evacuated from the pleural space by positive expiratory pressure and prevents air from being drawn back in.¹

3. A vent – an air outlet port that allows air to escape from the collection canister as air or fluid is drained from the pleural space. The outlet port can be attached to low-pressure suction to expedite evacuation of air and fluid.^{1-3,7}
- Chest drainage systems should never be raised above chest height and if tipped over, should be turned upright immediately.^{1,4,7}

Chest Drain Observations

- Frequency of core physiological observations (T, BP, pulse, respiratory rate, SpO₂, and conscious state) and chest drain observations are dependent upon the patient's clinical condition and medical orders. At a minimum, they should be performed 4- hourly whilst the chest drain is insitu.^{1,4}
- Chest drain observations include **swinging, bubbling, the amount, and type of drainage and suction level**^{1,2,4}, and are recorded on the Chest Drain Observation Chart.¹ The chest drainage system should also be assessed for **patency**.^{1,4†}

Swinging (Oscillation)

Ask patient to take a deep breath in and out. If a swing is present, fluid will rise up the straw inside the chest drainage canister on inspiration and fall on expiration.^{1,4}

Absence of a swing may occur if:

- The chest drain is occluded^{1,4,7}
- The chest drain is no longer within the pleural space^{1,4,7}
- The lung has fully expanded^{1,7}
- The chest drain is on suction^{1,4}

Bubbling

Ask patient to take a deep breath and cough.⁴ Bubbling in chest drainage canister indicates that air is being removed from pleural space and will occur with a pneumothorax, following lung surgery if there is a continuing air leak from the lung or if the patient is receiving positive pressure ventilation or CPAP.¹

A large volume air leak may indicate a broncho-pleural fistula.^{1,3}

Continuous bubbling on inspiration and expiration may indicate a leak in the system, a faulty connection, or an air leak at skin incision. Check connections are secure.^{1,4}

If bubbling suddenly ceases, ensure tubing or chest drain has not kinked, become blocked, or disconnected.^{1,4}

Drainage

Monitor at least every 4 hours and mark level of drainage on side of drainage canister. Document volume and type of drainage⁴, on Chest Drain Observation Chart and Fluid Balance Chart if in use.¹

If drainage increases significantly in amount or colour intensity, notify medical staff.^{1,4†}

- Any drainage of greater than 100mL of blood within 2 hours of insertion or post-surgery is significant and must be reported to the Medical Officer.¹ (excludes drains inserted for drainage of a pleural effusion).^{1,3}

Rapid drainage of effusions may result in re-expansion pulmonary oedema.² Typical signs are shoulder tip pain, coughing, sudden drop in blood pressure, and/or oxygen saturation, and increased respiratory rate and distress.¹

Large effusions should be drained incrementally, draining no more than 1.5L in the first hour.^{1,3} Clamps are used to control the rate of drainage. Stop drainage if the patient develops discomfort, persistent cough, hypotension or vasovagal symptoms (e.g. pallor, nausea, light-headedness, blurred vision, fainting).^{1,3}

If drainage suddenly ceases, ensure tubing or chest drain has not kinked.^{4,5,7}

Chest Drain Patency

Monitor tubing assessing from insertion site to the drainage canister ^{2,4} every 4 hours for:

- Kinks in tubing or chest drain^{1,5, 8}
- Presence of clots or debris⁵ – if present, gently tap or gently pinch the tubing just around the obstruction to move it into the drainage canister. Consider changing chest drain tubing set if unable to mobilise debris.¹
- Loosening of connections^{1,4, 7}
- Ensure air outlet port on drainage canister is unobstructed and vented to air when suction not connected.^{1,2,7-} The green ventilation cap should be inserted into the outlet port to prevent obstruction.⁷

Stripping or milking the tubing should not be performed as it creates high negative pressures, which can damage lung tissue.²

If chest drain blockage is suspected, change the chest drainage system⁷, monitor patient's clinical condition closely and seek immediate assistance from Clinical Nurse Specialist or Medical Officer.

Pain Management and Activity

- Assess pain scores and administer analgesia as prescribed to ensure patient's pain is well controlled.^{4,7} The patient should be able to deep breathe, cough and mobilise with minimal pain.¹
- Assist patient to reposition regularly. Position patient with head of bed elevated 30-45° when supine in bed. Encourage deep breathing and coughing to promote drainage and re-expansion of lung.^{1,2,}
- Encourage the patient to mobilise several times a day including sitting out of bed.² Refer to Physiotherapy if mobility assistance is required.¹
- Failure to adequately control pain may lead to decreased tidal volumes with subsequent hypoventilation, with risk of atelectasis and pneumonia. Observe the patient for clinical signs of infection. If present, notify Medical Officer.^{2,9-11}

Transporting Patient

- **Do not** clamp a Chest Drain for transport.^{1,4}
- A patient with a Chest Drain connected to an underwater seal drainage system must have a nurse escort familiar with chest drain management when leaving ward area.¹
- Ensure the canister is safely secured, remains upright and below the level of the patient's chest^{1,3,4}
- Ensure 2 x 18cm tubing clamps remain with the patient during transportation (large and small bore chest drains in the absence of 2 or 3 way tap).¹

Clamping of Chest Drains

- Clamping chest drains may result in a tension pneumothorax or surgical emphysema.^{2,3,}
- Clamping prior to drain removal is generally unnecessary.³
- Chest drains should only be clamped in carefully considered circumstances:
 - Briefly when replacing the chest drainage canister or chest drainage system^{1,2,}

- For short periods to drain collected fluid from the drain system tubing.¹
- At the request of the Medical Officer to control drainage rate of large pleural effusions.^{1,2}
- Briefly in the event it is not possible to maintain the chest drainage system below the patient's chest during a procedure such as bed to bed transfer.^{1,3}
- At the request of the Medical Officer to detect small air leaks not immediately obvious at the bedside.¹

- Patients undergoing a Clamping Trial must remain in the clinical area and be closely observed for signs of respiratory distress for the duration of the clamping trial. Immediately unclamp the chest drain if signs of clinical deterioration and inform the Medical Officer.^{2,3,}

Wound Management

- Remove post-insertion dressing after 24 hours.
- Inspect the insertion site for signs of infection, subcutaneous emphysema, or purulent pleural fluid. If present, notify the Medical Officer.^{1,2,4}
- A wound swab should be taken if any signs of clinical infection are present.^{7, 10†}
- If there is a significant air or fluid leak from the insertion site, ensure the patient is reviewed by Medical Officer in respect to chest drain placement and suture placement.⁷
- Minor air leaks from the insertion site may require an airtight dressing. Paraffin impregnated gauze wrapped around the chest drain at the insertion site and covered with a fenestrated dressing may assist in reducing the air leak.
- Ensure chest drain is secured to chest wall using adhesive tape to prevent kinking and pulling. (Tape around chest tube then onto patient's skin to avoid direct tube/skin contact).^{1,3}

Large bore chest drain:

- Using an aseptic non touch technique, clean drain sites every 24 hours and PRN with Chlorhexidine 2% in Alcohol 70% solution, allow to dry and apply a fenestrated dressing if sites are oozing and secure with tape.¹³ Leave site open if no ooze present. †

Small bore chest drain:

- Using an aseptic non touch technique, clean drain sites every 48 hours and PRN with Chlorhexidine 2% in Alcohol 70% solution, allow to dry and apply a transparent semi-permeable dressing.^{1,10, 13,}

Indwelling Pleural Catheters:

- See Section 8 (page 26).

Troubleshooting

i. Accidental Disconnection of Chest Drainage System from Chest Drain

If chest drain was previously draining air or both air and fluid (pneumothorax or haemopneumothorax):

- Do not clamp
- Cover end of chest drain with sterile towel or gauze.
- Call assistance to obtain and setup a new chest drainage system and connect immediately.^{1,12}

If chest drain was previously draining only fluid (pleural effusion):

- Cross-clamp the chest drain closer to the patient than the disconnection.
- Cover end of chest drain with sterile towel or gauze.
- Prepare and connect new chest drainage system immediately.
- Unclamp immediately.^{1,12}

ii. Accidental Dislodgement of Chest Drain

Accidental removal of Chest Drain is a medical emergency – call for assistance immediately

- Place a sterile dressing over insertion site using a non-touch technique. Place a paraffin impregnated dressing over the insertion site or alternatively remove the paper backing from a sterile gauze pack and place the gauze directly over the insertion site with plastic covering still in place.
- Tape 3 sides of the dressing ONLY leaving the inferior edge untaped to create an external flap valve.²
- Notify the Medical Officer immediately.
- Gather equipment and prepare for reinsertion of chest drain.
- The replacement chest drain should be inserted through a different insertion site.^{1,7}

iii. Chest Drainage System Knocked Over

If tipped over, the drainage system should be turned upright immediately.⁷

Check the underwater seal remains intact – the distal end of the straw should be immersed in at least 2cm of water.^{1,7}

iv. Surgical Emphysema

Surgical emphysema is the presence of air under the subcutaneous layer of the skin and is characterised by the feeling of crackling or rice bubbles on palpation.¹

Surgical emphysema starts at the insertion site and can spread.¹

Report developing or increasing surgical emphysema to the Medical Officer.¹

May be treated conservatively, or by applying suction or inserting a new chest drain.¹

Severe cases of surgical emphysema may cause changes in the patients' voice and facial appearance. Reassure patients and carers that it will not cause upper airway obstruction.¹

v. Excessive Bubbling and Frothing of Drainage Fluid

Rocket™ chest drainage canisters are lined with an antifoaming agent, which significantly reduces foaming that may occur as air bubbles through the canister contents.^{7, 15,}

1mL of Simethicone 10% liquid can also be added to the canister to reduce foam formation.¹

Excessive bubbling may indicate a large pleural leak. Continuous bubbling on inspiration and expiration may indicate a leak in the system, faulty connection, or air leak at skin incision.^{1,4}

Check all connections are secure^{1,7}

Inspect the insertion site:

- If there is a significant air leak from the insertion site, a Medical Officer should review chest drain placement and consider additional suture placement.
- Paraffin impregnated gauze may be wrapped around the chest drain at the insertion site and covered with a fenestrated dressing to reduce minor

insertion site air leaks.¹

If the chest drain is on suction, an overflow system can be used if excessive bubbling or foaming is present and likely to track up the suction tubing (see Section 5).¹

If the low-pressure suction regulator becomes contaminated with pleural fluid, the complete unit must be replaced. Place in sealed biohazard bag and send to Medical Technology and Physics for cleaning. †

2.0 Insertion and Post Insertion Management

2.1 Medical Requirements

Chest drain insertion requires written consent as per North Metropolitan Health Service Consent to Treatment Policy, and in compliance with the WA Health Consent to Treatment Policy (MP 0175/22)

All persons involved in the insertion and management of chest tubes or drains should be adequately trained and supervised.^{1,6} Outside of the Emergency Department or Operating Theatre, the procedure must only be performed by a Respiratory, Cardiothoracic or Intensive Care physician. ‡

Pleural procedures should not take place out of hours except in an emergency.^{1,3} Ultrasound guidance is recommended for drain insertions for pleural effusion.^{1,3,8} Coagulopathy should be excluded or corrected prior to attempting the procedure.^{1,3,6}

Consider IV access prior to the procedure. The procedure and drainage of fluid may result in hypotension, necessitating fluid resuscitation or replacement.¹

IV analgesia and/or sedation may be prescribed and administered by Medical Officer prior to insertion.^{1,3} If sedation is being used, the procedure should be performed with appropriate monitoring and resuscitation equipment immediately available, and supplemental oxygen should also be considered.⁶

A Chest X-ray is required post insertion to check Chest Drain placement, for the development of post insertion pneumothorax and to assess the need for suction.^{1,6} Confirmation of placement should be documented by Medical Officer in Inpatient notes.³

2.2 Safety Information

Tubing clamps (2 per tube) must remain with the patient at all times. However, drains should only be clamped in certain circumstances by staff experienced in the management of chest drains.^{1,4} Small bore chest drains with a 2- or 3-way tap are excluded.

Rapid drainage of pleural effusions may cause re-expansion pulmonary oedema.^{1,3} Typical signs of this are shoulder tip pain, coughing, sudden drop in blood pressure, and/or oxygen saturation, and increased respiratory rate and distress.¹

Large effusions should be drained incrementally, draining no more than 1.5L in the first hour.^{1,3} Clamps are used to control the rate of drainage. Stop drainage if the patient develops discomfort, persistent cough, hypotension, or vasovagal symptoms.^{1,3}

2.3 Nursing Management

Clip area of insertion with surgical clippers, if necessary.¹⁹

Explain procedure to patient including the need to refrain from coughing during procedure.

All fluid loss should be recorded accurately on Chest Drain Observation Chart and Fluid Balance Chart if in use. †

Procedural Information

Requirements (for all types of drain insertion)

- Large Trolley Fenestrated Drape
- Sterile Gown, Sterile Gloves, Protective Eye wear and Surgical Mask^{1,6}
- Dressing Pack / Catheter Pack
- Suture material 1.0 - 2-0 silk or prolene^{1, 17}
- 1 x 10ml syringe
- 1 x 50ml luer lock syringe (only if pleural fluid sample required)
- 1 x Blunt drawing-up needle
- 1 x 22G 1" needle
- 1 x 21G 2" needle
- 1 x 26G needle
- 2 x 5ml ampoules of Lignocaine 1%⁶
- Chlorhexidine 2% in 70% Alcohol or Povidine Iodine 10% solution
- 2 x 18cm Tubing Clamps per chest drain (required for all chest drains in the absence of a 2 or 3 way tap and must be retained at the bedside with the patient post procedure)
- Chest Drainage Canister
- 500 ml Sterile Water for Irrigation
- Chest Drain Tubing Set **(2-way tubing set available for use when two drains insitu)**
- Low Pressure Suction Unit with Duplex suction adapter (if suction requested)
- Appropriate lengths of adhesive tape e.g., Elastoplast™, Fixomul™
- Pulse Oximeter^{1,3,17}

Additional Equipment for Large bore insertion (Blunt dissection) †

- Chest drain of size requested by inserting physician
- Disposable Nasal speculum
- Disposable Scalpel blade No 11
- Surgical Minor Procedure Pack or Chest Drainage Insertion Set
- Fenestrated foam dressing¹

Additional Equipment for Small bore insertion (Seldinger technique)

- Chest Drain Insertion Set in size requested by inserting physician
- Clear occlusive dressing.^{1,6}

Additional Equipment (IPC insertion)

- IPC peritoneal or pleural insertion set
- Clear occlusive dressing.^{1,6}
- Fenestrated foam dressing†

Procedure (Nurse Assisting Medical Officer)

1. Medical Officer to explain procedure to patient and obtain consent⁶
2. Perform hand hygiene.
3. Clean the trolley using a combined detergent /disinfectant wipe, allow to dry.
4. Gather equipment for procedure.
5. Perform hand hygiene.
6. Perform and record core observations¹
7. Assist patient into required position³

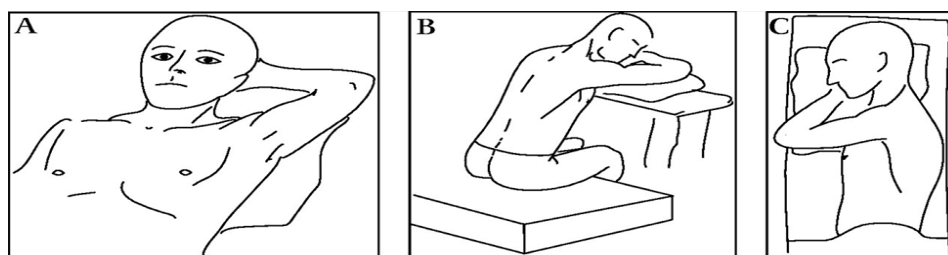


Figure 1 Common patient positions for chest drain insertion³:

- A) Semi-reclined with hand behind head.
- B) Sitting up leaning over a table with padding.
- C) Lateral decubitus position.

8. Attach the patient to pulse oximeter, observing for deterioration in SpO₂ throughout the procedure.^{1,17}
9. Perform hand hygiene. Perform hand hygiene.
10. Assemble equipment and assist Medical Officer maintaining a sterile field
11. Using aseptic technique, prepare the chest drainage system
12. Fill canister to prime level (zero mark) with 500mls sterile water.^{1,7}
13. Take care not to contaminate ends of tubing set when removing from the packaging as there are no protective covers on the ends. Maintain sterility of the end of drainage tubing by keeping it contained within plastic packaging.^{1,7,16}
14. Insert straw into canister ensuring the end of the straw is approximately 2 cm below the water level.^{1,7}
15. Write the time and date at the zero mark on the gauge on the side of the canister.^{1,7} (see Section 3)
16. Place the chest drainage system below the level of patient's chest.¹
17. Once chest drain is inserted, pass end of chest drain tubing to Medical Officer for connection to the chest drain. If chest drain is placed to treat a pleural effusion, re-expansion pulmonary oedema from rapid fluid drainage should be prevented by allowing a maximum of 1L of pleural fluid to drain at time of insertion. The chest drain should then be clamped, and further fluid drainage should be managed incrementally at hourly intervals if core observations are stable.¹ (see below)
18. Ensure all connections are secure and the tubing is not kinked.
19. Once drain fixation has been achieved with an anchoring suture, apply a fenestrated dressing (large bore) or a clear occlusive dressing (small bore or IPC)
20. Secure chest drain to chest wall using adhesive tape to prevent kinking and pulling.¹ (Tape around chest tube as below³ then onto patient's skin to avoid direct tube/skin contact¹).

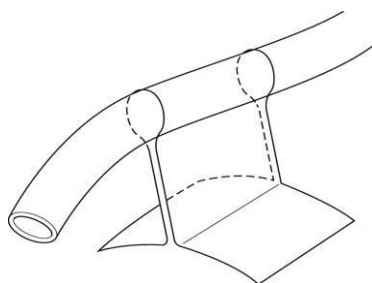


Figure 2: The omental tapetechnique³

21. Hang 2 x18 cm Tubing Clamps from the chest drainage system. These clamps should remain with chest drain at all times.^{1,17†}
22. A Chest x-ray should be performed immediately after insertion of chest drain to confirm adequate placement. Confirmation of placement should be documented by the Medical Officer in inpatient notes.¹
23. Whilst the Medical Officer removes sharps, waste and cleans equipment, commence

post insertion assessment and management.

POST INSERTION MANAGEMENT

1. Perform and record core physiological observations (T, BP, pulse, respiratory rate and SpO₂, conscious state) every 30 minutes for the first hour, then continue 4 hourly observations whilst chest drain is insitu. More frequent observations should be considered if sedation used, or if clinically indicated.^{1,3,4,20}
2. Perform chest drain observations (swinging, bubbling the amount and type of drainage and suction level) every 30 minutes for the first hour, then continue 4 hourly observations whilst chest drain is insitu.^{1,3,4} Record on Chest Drain Observation Chart (and Fluid Balance Chart if in use).^{1†}
3. Observe the patient for:
 - a) Deterioration in clinical condition¹
 - b) Tension Pneumothorax (e.g. deviation of the trachea, shortness of breath, tachycardia, hypotension)¹
 - c) Haemorrhage from site or excessive drainage in drainage bottle^{1,3}
 - d) Any drainage of >100mL of blood within 2 hours post insertion is significant must be reported to the Medical Officer. (N.B. excludes drains inserted for drainage of pleural effusion).^{1,3}
 - e) Pain – assess pain scores with observations^{1,4}
4. Large effusions should be drained incrementally, draining a maximum of 1.5L in the first hour. Greater drainage may lead to re-expansion pulmonary oedema.^{1,3} Clamp the drain following 1L drainage. Reassess core observations after 1 hour and if stable release clamps to allow a further 500mls to drain. Continue this process draining 500mls in each subsequent hour until drainage has slowed to < 500mls/hour then leave drain unclamped.¹
5. Stop drainage if the patient develops discomfort, persistent cough, hypotension or vasovagal symptoms.^{1,3} Escalate changes in core observations as per Observation Response Chart (MR 855).[†]
6. Educate the patient advising them of their responsibilities regarding care of the chest drain, including the importance of:
 - Maintaining the drainage system in an upright position and below the chest level especially when mobilising.¹
 - Requesting adequate analgesia to allow effective deep breathing & coughing and to permit mobilisation.¹
 - Avoiding kinking or obstruction of the chest drain tubing.¹
 - Requesting assistance to mobilise if chest drain is on suction as suction tubing can pose a tripping hazard to patient and others.¹
 - Remaining within the clinical area whilst chest drain is insitu. A nurse escort should accompany patient outside of the ward area.^{1,3 †}

3.0 Set up and Changing of Chest Drainage Systems

3.1 Safety Information

If the amount of fluid drainage is significant, the capacity for further air or fluid drainage may be impeded by the increasing hydrostatic pressure in the drainage collection canister. Monitor and change chest drainage canister when the fluid level reaches 1400ml (1000mL if drain is on suction). If the patient's respiratory status is worsening, consider changing the drainage bottle sooner.[†]

Suction must be turned off before clamping the Chest Drain to change the drainage canister or tubing set as an increase in negative pressure will be applied to lung tissue when the Chest Drain is unclamped.

3.2 Nursing Management

Change the drainage system when:

- a) Fluid level reaches 1400mL or 1000mL if drain is on suction.[†]
- b) Chest drain tubing or drainage canister is damaged ⁸
- c) Substantial exudate is adhering to the tubing.¹

3.3 Procedural Information

Requirements

- Chest Drainage Canister
- Chest Drain Tubing Set (if requires changing)
- 18cm Tubing Clamps x 2 per chest drain
- 500 ml Sterile Water for Irrigation
- 70% Alcohol large wipe
- Personal Protective Equipment (PPE)
- Wet strength bag

Preparation

Changing Chest Drainage Canister only

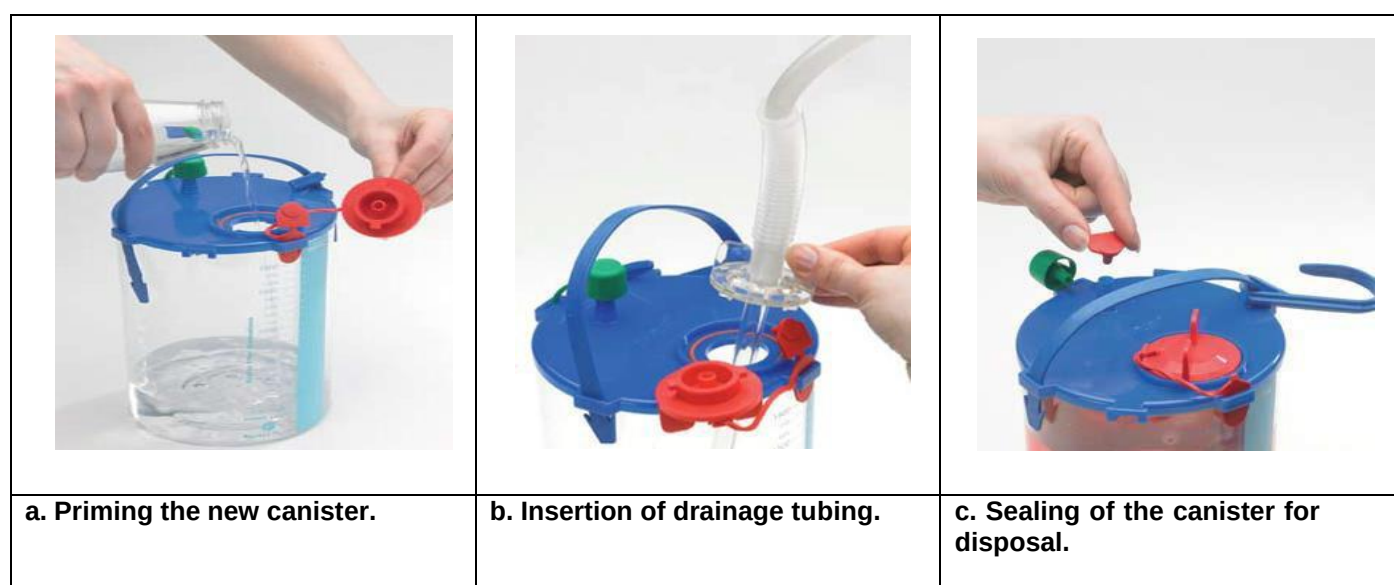
1. Perform hand hygiene.
2. Gather equipment for the procedure.
3. Perform hand hygiene and don PPE as per standard precautions.
4. Rotate red cap to open canister.
5. Fill canister to prime level (zero mark) with 500mls sterile water.^{1,7,16,21} (see Figure 3a)
6. Write the time and date at the zero mark on the gauge on the side of the canister.^{1,7,16} and perform hand hygiene.
7. If chest drain is attached to suction, turn off and disconnect suction tubing from the airoutlet port.⁷
8. Clamp the Chest Drain by cross clamping with two 18cm tubing Clamps or close 2 or 3way tap.¹
9. Transfer existing chest drain tubing set into newly prepared drainage canister using an aseptic non touch technique.⁷
10. Ensure the connections are secure.^{4,7}
11. Unclamp the chest drain, check for swinging, bubbling and drainage.
12. Reconnect suction if required. If not on suction, cover the air outlet port with the green ventilation cap.⁷
13. Dispose of waste. To discard old drainage canister, remove tubing set, and seal canister and air outlet port with the red caps attached to canister.^{7,16} (see Figure 3c) Discard canister with contents into 'waste for incineration' bin.¹
14. Remove PPE and perform hand hygiene.
15. Document any further drainage since last measured, and canister or drainage system change on Chest Drain Observation Chart.^{1,8}

Changing Chest Drain Tubing and Drainage Canister

1. Complete steps 1-5 as above
2. Take care not to contaminate ends of tubing set when removing from the packaging, as there are no protective covers on the ends. Maintain sterility of the end of drainage tubing by keeping it contained within plastic packaging.^{7,8,16}
3. Insert straw into canister ensuring the end of the straw is approximately 2 cm below the water level.^{7,16} (see Figure 3b)
4. Write the time and date at the zero mark on the gauge on the side of the canister.^{7,16} and perform hand hygiene.
5. If chest drain is attached to suction, turn off and disconnect suction tubing from the airoutlet port.⁷

6. Clamp the chest drain by cross clamping with two 18cm Tubing Clamps or close 2 or 3 way tap.¹
7. Perform hand hygiene and don PPE.
8. Place absorbent drape under the connection and the catheter.
9. Clean connection between chest drain tubing set and chest drain using 70% Alcohol large wipe and allow to air dry for 30 seconds.
10. Using an aseptic non touch technique, disconnect old chest drain tubing set and insert new chest drain tubing into chest drain.
11. Ensure connections are secure.
12. Unclamp chest drain, check for swinging, bubbling and drainage.
13. Reconnect suction if required. If not on suction, cover the air outlet port with the green ventilation cap.^{7,16,21}
14. Dispose of waste. To discard old drainage canister, remove chest drain tubing set and seal canister and air outlet port with the red caps attached to canister.^{7,9} (see Figure 3c) Discard bottle with contents into 'waste for incineration' bin.¹ Place old chest drain tubing set into wet strength bag and discard in general waste. †
15. Remove PPE and perform hand hygiene.
16. Document any further drainage since last measured, and canister or drainage system change on Chest Drain Observation Chart .

Figure 3: Canister Management²¹ (note: non-sterile gloves must be worn)



4.0 Management of Chest Drain Suction

4.1 Medical Requirements

A written or verbal medical order is required for the application or cessation of suction to a chest drain.¹

4.2 Safety Information

A **low-pressure thoracic suction regulator** (see Figures 4a and 4b) **MUST** be used, and suction pressure must not exceed -40cmH₂O or -5kpa.¹⁻³ A filter is recommended for use with the Ohio unit (Figure 4b)

If the bed space has only one wall suction outlet, a **Clements Duplex Suction Adaptor** (see Figure 4c) **MUST** be used to allow connection of both the low-pressure thoracic suction regulator and the emergency suction unit to the single wall suction outlet (see Figure 4d below).²³

Figure 4: Suction Regulator Type and Set-up²⁴

Check the low-pressure thoracic suction regulator is set to the prescribed suction level and document level on Chest Drain Observation Chart or document UWS if no suction prescribed.^{1†}

When suction is discontinued, the suction tubing must be disconnected from the air outlet port on the drainage canister immediately after turning the suction off to allow the system to be vented to air.^{7,16} Failure to do so may lead to a tension pneumothorax.^{1,2}

Rocket™ chest drain canisters are lined with an antifoaming agent which significantly

reduces foaming that can occur as air bubbles through the canister contents.^{7,15,16}

4.3 Procedural Information

Application of Suction

REQUIREMENTS

- 30m pack of suction tubing (consider patient condition and location and cut to length)
- Low-pressure thoracic suction regulator
- Clements duplex suction adaptor

PROCEDURE

1. Perform hand hygiene
2. Connect low-pressure thoracic suction regulator to wall suction outlet (use a Clements duplex suction adaptor if the bed space has only one wall suction outlet)
3. Connect the outer end of suction tubing to the low-pressure thoracic suction regulator.
4. Remove green ventilation cap from the air outlet port and attach the inner end of suction tubing.⁷
5. Dial required suction level slowly, assessing patient for pain.
6. Perform hand hygiene[†].
7. Check the low-pressure thoracic suction regulator is set to the prescribed suction level with each set of chest drain observations and document on Chest Drain Observation Chart.¹

Discontinuation of Suction

PROCEDURE

1. Perform hand hygiene
2. Turn suction off at the low-pressure thoracic suction regulator and disconnect the suction tubing from the air outlet port on the canister.
3. Replace green ventilation cap in the air outlet port.⁷
4. Discard suction tubing in general waste bin.
5. Perform hand hygiene.[†]

5.0 Use of and Overflow System

1.1. Safety Information

An overflow system can be used if excessive bubbling or foaming is present and likely to track up the suction tubing.¹

Rocket [™] chest drain canisters have an antifoaming agent lining them which significantly reduces foaming that can occur as air bubbles through the canister contents.^{7,15,16}

If the low-pressure suction regulator becomes contaminated with pleural fluid (the filter will change colour i.e., white to blue), the unit must be sealed in a biohazard bag, replaced, and sent to Medical Technology and Physics for cleaning and checking. Inform Engineering of the potential contamination of the system[†]

5.1 Nursing Management

The overflow chest drain bottle does not need priming with water, it must remain empty.[†]

Procedural Information

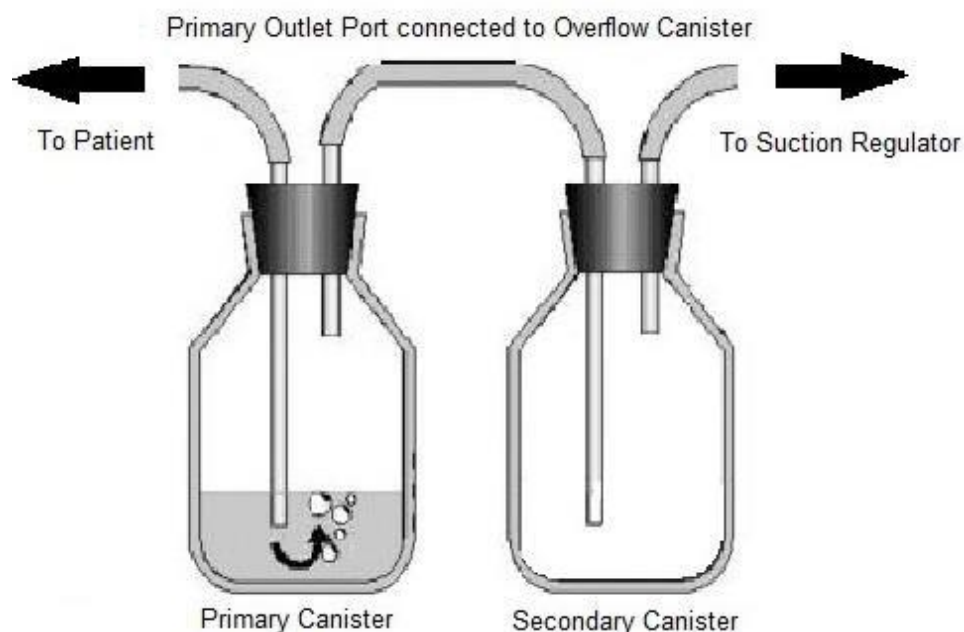
Requirements

- Chest drainage canister (overflow canister)
- Single chest drain tubing set

Procedure

1. Perform hand hygiene.
2. Gather equipment for the procedure.
3. Perform hand hygiene.
4. Insert new tubing set into empty overflow canister. **Do not prime with water.**
5. Turn off suction and disconnect suction tubing from the outlet port of the primary chest drainage canister.
6. Cut the soft tubing of the new chest drain tubing set to required length (~40cm) and connect to outflow port on primary drainage canister. The canisters should now be connected. (see Diagram 1: Overflow System)
7. Reconnect suction tubing from suction regulator to the outflow port on the overflow canister and recommence suction at the prescribed level.
8. Perform hand hygiene. †

Diagram 1: Overflow System²⁵



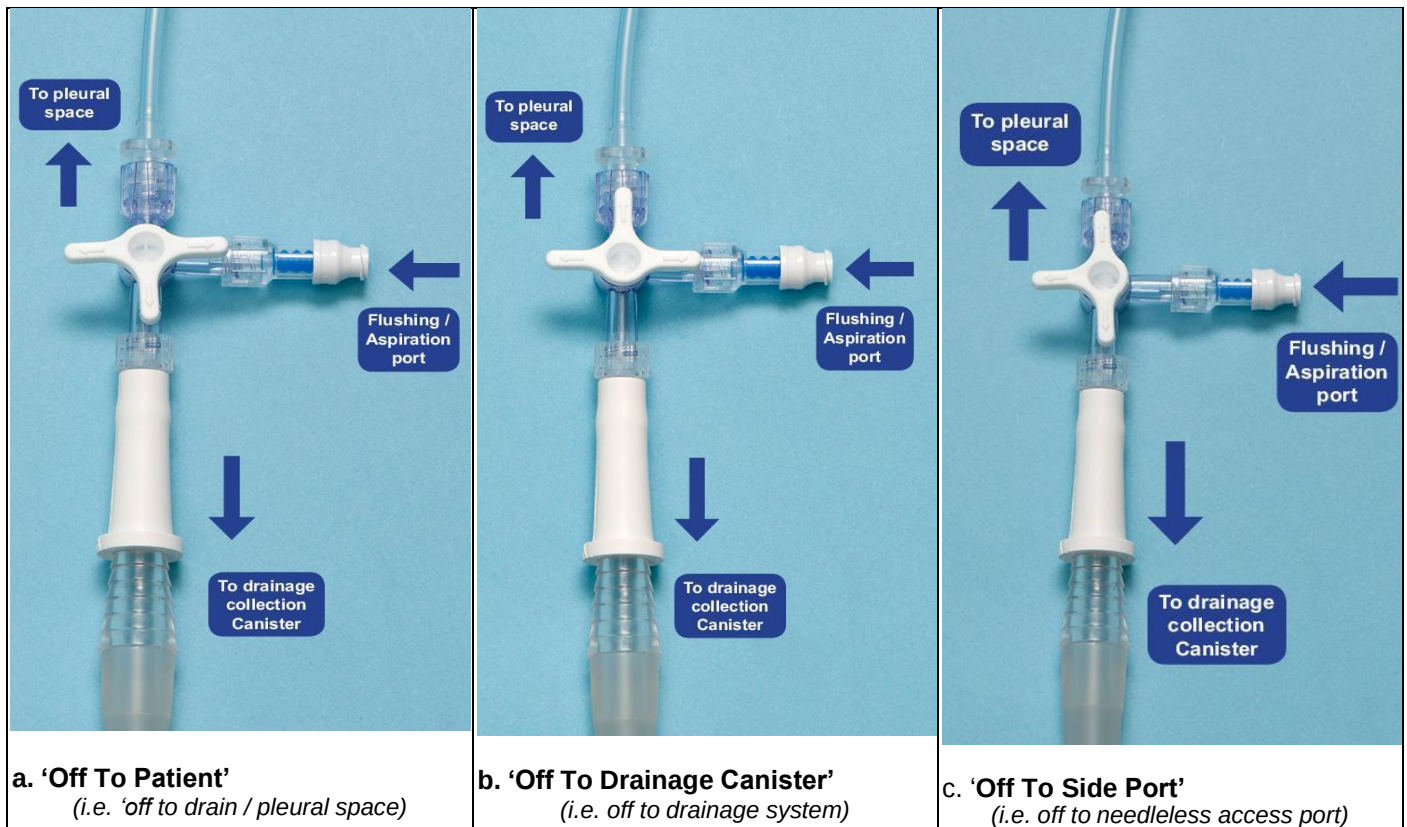
6.0 Flushing of Small Bore Chest Drains

6.1 Medical Requirements

0.9% Sodium chloride flushes for unblocking or flushing a small bore catheter must be prescribed on the WA Hospital Medication Chart (WAHMC) by a Medical Officer and administered by a Medical Officer or Registered Nurse.¹

6.2 Safety Information

A small bore chest drain may have a 3-way tap inserted into the circuit to allow flushing or administration of other prescribed medications.¹ It can be turned one of 3 ways: 'Off to the Patient', 'Off to the Drainage Canister', or, 'Off to the side Port' (see Figure 5 below).

Figure 5: Positions of the Small Bore Chest Drain Tap²⁶

When the small-bore chest drain is connected to the drainage system, the preferred 3-way tap position is '*off to the side port*'. This is to ensure drain patency and to prevent air entering the pleural space should the needleless port become accidentally disconnected.

†

Ensure 0.9% Sodium Chloride flushes are ordered on the WA Hospital Medication Chart (WAHMC).[†]

Procedural Information

Requirements

- Personal Protective Equipment (PPE)
- 1 x sterile 20ml luer lock syringes with 20 ml 0.9 % sodium chloride for irrigation.
- 1 x 70% alcohol large wipe

Procedure

1. Perform hand hygiene and apply PPE.
2. Scrub the needleless access port for 30 seconds with a 70%alcohol large wipe. Use different areas of the wipe.
3. Allow to air dry for 30 seconds.
4. Attach syringe to needleless port.
5. Turn 3-way tap '**off to drainage canister**' (see Figure 5b)
6. Gently flush the pleural drain with 20mls of sodium chloride 0.9% (Figure 6)

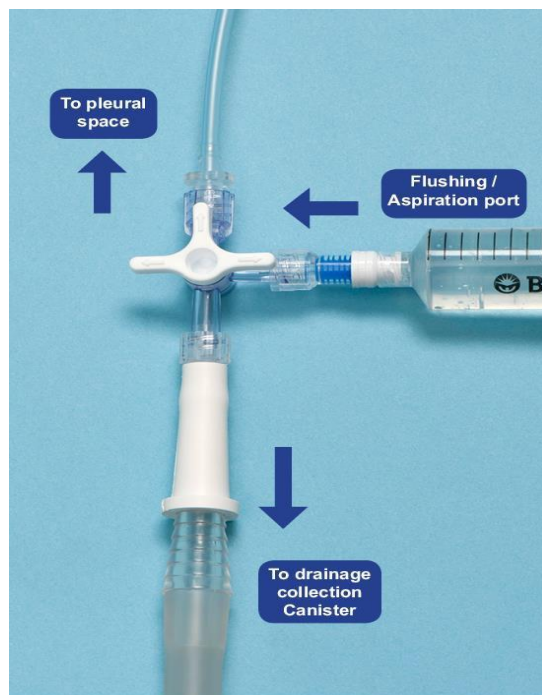


Figure 6: Flushing of small bore chest drain²⁶

7. Prior to disconnecting syringe turn 3-way tap '**off to side port**' (see Figure 5c)
8. Disconnect syringe.
9. Remove gloves and perform hand hygiene.
10. Document the flushing with 0.9% Sodium Chloride on the WAHMC and the additional 20ml of 0.9% Sodium Chloride on the Chest Drain Observation Chart. [†]

7.0 Removal of Chest Drain

7.1 Medical Requirements

Review patient condition and coagulation status **prior to ordering chest drain removal** particularly if the chest drain was inserted using a Seldinger technique.¹

The decision for chest drain removal must be documented in the inpatient notes.¹

A post removal chest X-ray is a decision for the treating physician but is recommended within 2 –4 hours post removal of large bore chest drains.¹

Routine chest x-ray is no longer required post removal of small-bore drains in the asymptomatic patient.^{27,28}

7.2 Safety Information

This guideline refers to the removal of an intercostal catheter / chest drain **not** an indwelling pleural catheter (IPC). Removal of an IPC is a surgical procedure performed by Respiratory Physicians only. ^{1 †}

Two nurses, one whom has experience in this procedure are required for large bore chest drain removal.^{1,29}

If a closing suture has been inserted for a small bore chest drain, follow large bore chest drain removal steps with two nurses.[†]

Chest drains inserted in interventional radiology may have a locking mechanism, please refer to manufacturer guidelines for instruction on releasing the locking mechanism.

If strong resistance is felt during chest drain removal (especially if a locking mechanism is present), cease attempted removal and notify CNS, Interventional Radiology or Medical Officer.¹

7.3 Nursing Management

Leave dressing intact for 48 hours post chest drain removal.¹

Remove closing suture at 5 days post chest drain removal.[†]

Procedural Information

REQUIREMENTS

- Trolley
- Dressing Pack
- Chlorhexidine 2.0% in Alcohol 70% solution
- Stitch cutter
- Absorbent dressing (e.g., small Zetuvit™ or gauze)
- Paraffin impregnated gauze (large bore)
- Clear occlusive dressing (small bore with no closing suture in place)
- Adhesive tape e.g., Elastoplast™ or Fixomul™ (large bore)
- Personal Protective Equipment (PPE) including eye protection
- Plastic backed protective sheet
- Wet strength bag
- Device to release locking mechanism of small bore chest drains (as applicable)¹

PROCEDURE

1. Confirm physician orders for removal and ensure adequate analgesic cover 30 minutes prior to removal of Chest Drain.¹
2. Confirm if a second nurse is required for the procedure:
 - i. A second nurse **must assist** when removing large bore intercostal catheters (18-32Fr.).¹
 - ii. A second nurse is not usually required when removing small bore catheters and drains (12-16Fr.).
3. Perform hand hygiene.
4. Clean the trolley using a combined detergent /disinfectant wipe, allow to dry.
5. Gather equipment for the procedure.
6. If chest drain is attached to suction, turn off and disconnect suction tubing at canister.¹
7. Explain the procedure to the patient and advise they may experience a brief sensation of burning or pain as the drain is removed. Describe, demonstrate, and then ask patient to practice breath holding for removal. i.e. Take a deep breath in and hold it. If patient is unable to perform breath hold, time removal of drain with exhalation.¹
8. Perform hand hygiene and position patient in an upright position with head of bed elevated 35-40° to allow ease of access to the insertion site and place plastic backed protective sheet underneath patient adjacent to the insertion site.¹
9. Perform hand hygiene and prepare sterile field: open procedure pack using corners and add sterile equipment onto sterile field using a non touch technique.
10. Perform hand hygiene and don PPE (operator and assistant)
11. Prepare dressing; placing folded paraffin gauze onto centre of absorbent dressing.
12. Clean insertion site with 2% Chlorhexidine in Alcohol 70% solution and allow to dry.^{1,20,30}

Two Nurse Removal Procedure for Large Bore Intercostal Catheters (18-32 French)

13. Ascertain if a 'closing' suture is in place.
14. Remove anchor suture from insertion site whilst supporting the chest drain.¹

15. Loosely half tie closing suture.
16. Operator: cover insertion site with prepared occlusive dressing and request the patient to perform breath hold. Once breath is effectively held, remove the chest drain in a swift, fluid motion and apply firm pressure over insertion site.¹
17. Maintain pressure over site and instruct the patient to release their breath and breathe normally.¹
18. Assistant: tighten and re-tie the closing suture twice and trim excess suture material. Secure the dressing with 2-3 lengths of adhesive tape while the operator continues to apply firm pressure until dressing secured. Label with date and time of removal^{1,29}
19. Dispose of waste. To discard drainage canister, remove tubing set, and seal canister and air outlet port with the 2 red caps attached to canister.^{7,16} Discard canister with contents into 'waste for incineration' bin.¹ Place tubing set and dressing materials into wet strength bag and discard in general waste. Dispose of sharps appropriately.
20. Remove PPE and perform hand hygiene.
21. Clean the trolley using a combined detergent /disinfectant wipe, allow to dry and perform hand hygiene.^{1, †}
22. Record removal date, time and volume of drainage in inpatient notes and Chest Drain Observation Chart.[†]

Single Nurse Removal Procedure for Small Bore Catheters & Drains (12-16 French).

13. Remove the anchor suture from the insertion site / or release the locking mechanism whilst supporting the chest drain.¹
14. Cover insertion site with prepared dressing and request the patient to perform breath hold. Once breath is effectively held, remove the chest drain in a swift, fluid motion and apply firm pressure over insertion site.¹
15. Maintain pressure over site and instruct the patient to release their breath and breathe normally.¹
16. Apply clear occlusive dressing over the insertion site and label with date and time of removal.¹
17. Dispose of waste. To discard drainage canister, remove tubing set, and seal canister and air outlet port with the 2 red caps attached to canister.^{7,16} Discard canister with contents into 'waste for incineration' bin.¹ Place tubing set and dressing materials into wet strength bag and discard in general waste. Dispose of sharps appropriately.
18. Remove PPE and perform hand hygiene.
19. Clean the trolley using a combined detergent /disinfectant wipe, allow to dry and perform hand hygiene.^{1†}
20. Record removal date, time and volume of drainage in inpatient notes and Chest Drain Observation Chart.^{1†}

8.0 Indwelling Pleural Catheter (IPC)

8.1 Medical Requirements

The volume and frequency of drainage is to be advised by Respiratory Physician and documented in inpatient notes.

8.2 Nursing Management

Indwelling Pleural Catheters (IPC) are silicone tubes that are tunnelled and secured subcutaneously (with a pro-fibrotic cuff), and end with a one-way valve. They are inserted into the pleura to manage recurrent symptomatic pleural effusions.¹⁴

Pleural fluid is drained into an underwater seal system on insertion. Once it has been confirmed no air leak is present, subsequent drainage may be completed using a 'male' luer lock drainage collection bag.

IPC's can be 'capped' off when pleural drainage is not required and accessed with drain

specific connectors via a one way valve as required.⁵

Removal of an IPC is a surgical procedure **performed by Respiratory Physicians only**.

Procedural Information

ii. Accessing the 'Capped' Indwelling Pleural Catheter REQUIREMENTS

- Personal protective equipment
 - IPC drainage line (contains an IPC graduated connector)
 - IPC replacement valve cap
 - Chest Drainage Canister
 - 500 mL Sterile Water for Irrigation
 - Chest Drain Tubing Set
 - Dressing pack
 - 70% alcohol large wipe
 - Sterile scissors
- } **or** Male luer lock 2L collection bag

Procedure

1. Perform hand hygiene.
2. Clean the trolley using a combined detergent /disinfectant wipe, allow to dry.
3. Gather equipment for procedure.
4. Perform hand hygiene and prepare sterile field: open procedure pack using corners and add sterile equipment onto the sterile field using a non touch technique.
5. If using an under water seal (UWS) system;
 - attach the IPC graduated connector to IPC drainage line.
 - set up UWS system as per Section 3
 - cut off the graduated connector from end of chest drain tubing set with sterile scissors.
 - attach the chest drain tubing to the IPC graduated connector on IPC drainageline
6. If using Drainage bag attach male luer lock collection bag directly to the IPC drainage line.
7. Remove the packaging from IPC drainage line as per Figure 7. Close pinch clamp online.



Figure 7: IPC Access tip ^{31,32},

8. Remove IPC cap by gently rotating anti-clockwise and pulling away as per Figure 8, discard cap.



Figure 8: Rotate anti-clockwise to remove IPC cap³²

9. Using a non-touch technique, scrub the one way valve for 30 seconds with a 70% alcohol wipe. Use different areas of the wipe. Allow to air dry for 30 seconds.
10. Gently insert the access tip of the IPC drainage line into the valve. It is normal to feel a little resistance as the one-way valve is breeched. Lock the two together by lining them up as shown in Figure 9 and rotating clockwise. You will hear a click.

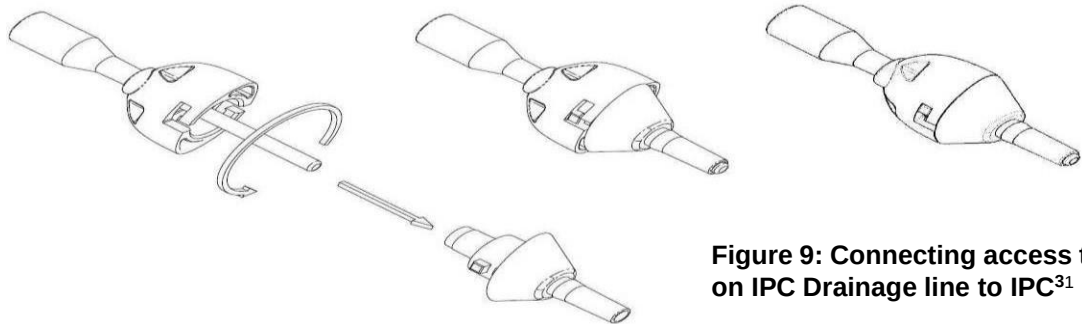


Figure 9: Connecting access tip on IPC Drainage line to IPC³¹

11. Attach chest drain tubing to IPC drainage line at the graduated connector and open pinch clamp.
12. Allow drainage as per advised volume, document and monitor as per general care instructions (Section 1).
13. Large pleural effusions should be drained incrementally, draining a maximum of 1.5L in the first hour. Greater drainage may lead to re-expansion pulmonary oedema. Clamp the drain with pinch clamp following 1L drainage. Reassess core observations after 1 hour and if stable release pinch clamp to allow a further 500mls to drain. Continue to drain 500mls in each subsequent hour until drainage has slowed to < 500mls/hour then leave drain unclamped.^{1-3,5}
14. Stop drainage if the patient develops discomfort, persistent cough, hypotension or vasovagal symptoms.³ Escalate changes in core observations as per Observation Response Chart.[†]
15. Remove PPE, and perform hand hygiene.

iii. 'Capping the IPC

After drainage is complete the IPC will need to be capped off.

Requirements

- Personal protective equipment
- IPC replacement valve cap
- 70% alcohol large wipe
- 2% Chlorhexidine 70% alcohol solution
- Fenestrated foam dressing
- Clear occlusive dressing
- Gauze
- Dressing pack
- Fixomul™

PROCEDURE

1. Perform hand hygiene and don PPE. Disconnect the drain from the IPC by gently rotating the drainage line anti clockwise and gently pulling out the access valve in a smooth motion. The drain itself has a one way valve which automatically closes once disconnected.
2. Using a non-touch technique, scrub the one way valve for 30 seconds with a 70% alcohol large wipe. Use different areas of the wipe. Allow to air dry for 30 seconds.
3. Place a new catheter valve cap over valve and rotate it clockwise until it clicks into position.
4. Clean around drain exit site with 2% Chlorhexidine 70% alcohol solution and allow to dry.

5. Place fenestrated foam dressing around drain and gently curl drain around foam dressing as Figure 10.



Figure 10: Securement and dressing of IPC³²

6. Cover with gauze to prevent the occlusive dressing sticking to the drain directly and then cover with occlusive dressing.
7. Protect edges of occlusive dressing with Fixomul™ to extend life of dressing.
8. Dispose of waste.
9. UWS system: To discard drainage canister, remove tubing set, and seal canister and air outlet port with the 2 red caps attached to canister. Discard canister with contents into 'waste for incineration' bin.¹ Place tubing set, IPC drainage line and dressing materials into wet strength bag and discard in general waste.
10. Male luer lock 2L collection bag: Empty collection bag into clean measuring container and dispose of contents appropriately. Place collection bag, IPC drainage line and dressing materials into wet strength bag and discard in general waste.
11. Remove PPE and perform hand hygiene.
12. Record drainage on Chest Drain Observation Chart (and Fluid Balance Chart if in use)
13. Change dressing weekly or PRN.
14. Educate patient to use shower protectors to prevent dressing becoming wet.

9.0 Pleural Aspiration

9.1 Medical Requirements

Recent chest x-ray or ultrasound equipment

9.2 Safety Information

Consider IV access prior to the procedure. The procedure and drainage of fluid may result in re- expansion pulmonary oedema.^{1,2}

IV analgesia and/or sedation may be prescribed and administered by Medical Officer prior to aspiration.³

9.3 Nursing Management

Assist with positioning of patient.³

Request patient to remain still and refrain from coughing.[†]

Procedural Information

Requirements

- Fenestrated Drape
- Sterile Gown Sterile Gloves
- Dressing Pack / Catheter Pack
- Personal Protective Equipment

- Chlorhexidine 2.0% in Alcohol 70% or Povidone Iodine 10%
- 2 x 5 ml 1% Lignocaine
- 1 x 20 ml syringe
- 1 x 50 ml syringe
- 1 x blunt Drawing up needle 1 x 21 G 2" needle
- 1 x 26 G ½" needle
- 1 x 14G 2" IV cannula or thoracentesis set
- 1 x 3 way tap
- 1 x male luer lock liquid collection bag
- Transparent Occlusive Dressing
- Cytology Bottles
- Sterile yellow screw top containers
- Blood Culture Bottles
- Arterial Blood Gas syringe
- Pulse oximeter

PROCEDURE

1. Perform hand hygiene.
2. Clean the trolley using a combined detergent /disinfectant wipe, allow to dry.
3. Gather equipment for procedure.
4. Perform hand hygiene.
5. Perform and record T, BP, pulse, respiratory rate and SpO2 and conscious state (core observations).
6. Monitor SpO2 throughout procedure.
7. Perform Hand hygiene.
8. Apply transparent occlusive dressing to aspiration site on removal of cannula using a non-touch technique.
9. Once pleural aspiration is complete; perform and record core observations (T, BP, pulse, respiratory rate and SpO2, conscious state) every 30 minutes for the first hour, then continue 4 hourly observations unless otherwise clinically indicated.
10. Record drainage volume on Fluid Balance Chart if in use or in the inpatient notes.
11. Don PPE and dispose of waste.
12. Remove PPE and perform hand hygiene.
13. Clean the trolley using a combined detergent /disinfectant wipe after use and perform hand hygiene†

Compliance, monitoring and evaluation

Related Documents

Legislation:

- Health Services Act 2016
- Work Health and Safety Act 2020
- Health Practitioner Regulation National Law (WA) Act 2010
- Nursing and Midwifery Board of Australia (NMBA) Code of Professional Conduct for Nurses 2018
-

Policy:

- WA Health Consent to Treatment Policy (MP 0175/22)
- NMHS Clinical Documentation Policy
- SCGOPHCG Consent to Treatment Policy
- SCGOPHCG Infection Prevention and Management Manual - Aseptic Technique Policy
- SCGOPHCG Hospital Policy No 232 Clinical Handover- Communicating for Safety.
- NMHS Policy Patient Identification & Procedure Matching
- OPH Hospital Policy Patient Identification
- SCGH Hospital Policy Patient Identification

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†**Consensus recommendation** by Chest Drain Management NPG Clinical Review Group.

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For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

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